Huawei Access Node MALL-ACC-03 Standard Operating Procedure for Routing Protocol Flap (BGP route churn, OSPF adjacency loss, ISIS LSP flooding)

Node: MALL-ACC-03 | **Vendor:** Huawei | **Severity:** Minor | **Alarm Code:** ALARM_6927

Description

The Huawei Access Node MALL-ACC-03 is a critical component of the Malleswaram Node, responsible for providing high-speed internet connectivity to the surrounding area. The failure domain of Routing Protocol Flap (BGP route churn, OSPF adjacency loss, ISIS LSP flooding) is a common issue that can occur on this specific role and vendor equipment due to the complex interplay of routing protocols and hardware mechanics. When the BGP route churn is triggered, it can lead to a cascade of events including OSPF adjacency loss and ISIS LSP flooding, resulting in the ALARM_6927. This is often caused by a combination of factors such as misconfiguration, hardware faults, or environmental issues. To mitigate this issue, it is essential to understand the underlying protocol mechanics and hardware architecture of the Huawei Access Node MALL-ACC-03.

Pre-requisite Checks

- Ensure that the Malleswaram Node is accessible and the Huawei Access Node MALL-ACC-03 is reachable via the command line interface (CLI).

- Verify that the redundant links are functioning correctly and there are no known issues with the physical infrastructure.

- Review the event logs and system alerts to identify any potential causes of the failure domain.

Potential Root Causes

Cause	Probability
Misconfigured BGP routing policy	High
Hardware fault in the Huawei Access Node MALL-ACC-03	Medium
Environmental issue affecting the router's performance	Low

Diagnostic Commands & Expected Outputs

```
Command: display bgp routing-policy, Expected Output: Display the BGP routing policy configuration, Description: This command will display the BGP routing policy configuration, including any misconfigured policies that may be causing the BGP route churn.

Command: display ospf neighbor, Expected Output: Display the OSPF neighbor information, Description: This command will display the OSPF
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neighbor information, including any issues with adjacency loss.

Command: display isis lsp, Expected Output: Display the ISIS LSP
information, Description: This command will display the ISIS LSP
information, including any issues with flooding.
```

Resolution Steps

Step 1: Step: Clear the BGP routing policy cache, Command: clear bgp routing-policy, Description: This command will clear the BGP routing policy cache and force the router to re-negotiate the BGP routes.

Step 2: Step: Check and clear the OSPF neighbor information, Command: clear ospf neighbor, Description: This command will check and clear the OSPF neighbor information, including any issues with adjacency loss.

Step 3: Step: Verify and clear the ISIS LSP information, Command: clear isis lsp, Description: This command will verify and clear the ISIS LSP information, including any issues with flooding.

Post-Check Verification

```
Command: display bgp routing-policy, Expected Output: Verify the BGP
routing policy configuration, Description: This command will verify
the BGP routing policy configuration, ensuring that the configuration
is correct and the BGP route churn has been resolved.

Command: display ospf neighbor, Expected Output: Verify the OSPF
neighbor information, Description: This command will verify the OSPF
neighbor information, ensuring that the adjacency loss has been
resolved.

Command: display isis lsp, Expected Output: Verify the ISIS LSP
information, Description: This command will verify the ISIS LSP
information, ensuring that the flooding has been resolved.
```

Roll-back Procedures

- Step: Revert the BGP routing policy configuration, Command: undo bgp routing-policy, Description: This command will revert the BGP routing policy configuration to its previous state, ensuring that the router returns to its previous configuration.

- Step: Revert the OSPF neighbor information, Command: undo ospf neighbor, Description: This command will revert the OSPF neighbor information to its previous state, ensuring that the adjacency loss is restored.

- Step: Revert the ISIS LSP information, Command: undo isis lsp, Description: This command will revert the ISIS LSP information to its previous state, ensuring that the flooding is restored.